

11

Eigenvalue Decomposition

The Eigenvalue Is a Matrix's "Own Value," the Eigenvector Is a Matrix's "Own Direction"

Eigenvalue decomposition helps us find a square matrix's eigenvectors and eigenvalues; an eigenvector can be understood as the matrix's "own direction," and an eigenvalue can be understood as its "own value." We'll soon see, in the next chapter, how eigenvalue decomposition is used in the principal component analysis algorithm.

Please pay special attention: the eigenvalue decomposition of a real symmetric matrix has a special name — "spectral decomposition." This is because matrices such as the Gram matrix, the covariance matrix, the linear correlation coefficient matrix, the adjacency matrix of an undirected graph, and the Laplacian matrix are all common examples of real symmetric matrices.

11.1 Eigenvalue Decomposition



Core skills covered in this section:

- ▶ An eigenvector is a nonzero vector whose direction remains unchanged under a square matrix's linear transformation.
- ▶ An eigenvalue represents the factor by which an eigenvector is scaled during the transformation.
- ▶ Solving eigenvalue decomposition by hand using the characteristic polynomial.
- ▶ All the eigenvectors for a given eigenvalue form a line through the origin.
- ▶ A square matrix is diagonalizable if and only if all of its eigenvectors are linearly independent.
- ▶ The eigendecomposition of a real symmetric matrix is called spectral decomposition.

This section introduces the basic concepts of eigenvalues and eigenvectors, and explains the geometric meaning and computation method of eigenvalue decomposition. In particular, the eigendecomposition of a real symmetric matrix can be understood as a "rotate → scale → rotate" transformation structure.

What Is an Eigenvalue? What Is an Eigenvector?

The word "eigen" in "Eigenvalue Decomposition" (EVD) comes from the German word "eigen," which means "own" or "inherent."

So the "eigenvalue" (also called the characteristic value) obtained from eigenvalue decomposition actually means an "inherent value" or "own value"; the "eigenvector" (also called the characteristic vector) means an "inherent vector" or "own direction."

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Taking the square matrix A as an example, eigenvalue decomposition means finding matrix A 's eigenvalue λ and the corresponding eigenvector v such that

$$A @ v = \lambda v$$

Eigenvector v , acted on by A , produces the vector λv .

This means that, as Figure 1 shows, under the action of A , the direction of the eigenvector v remains unchanged (either the same direction or the reverse). The eigenvalue λ determines the scaling, and the sign of λ determines whether the direction stays the same or reverses.

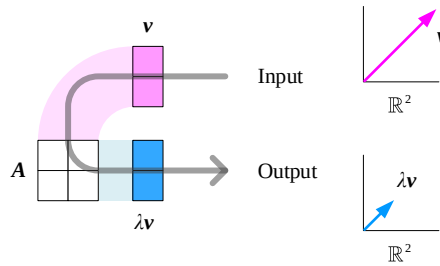


Figure 1. $Ax = y$, matrix A has shape 2×2

⚠ Note that eigenvectors are generally assumed to be nonzero, because $A0 = 0$ trivially means, for a linear map, that "the origin stays fixed" — not worth discussing. However, please pay special attention: the eigenvalue can be 0.

Let's look at an example

Given the 2×2 square matrix A

$$A = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

As Figure 2 shows, let's apply matrix A as a linear transformation to the unit vectors in the figure. As mentioned earlier in this book, if a unit vector in the plane starts at the origin, its endpoint lies on the unit circle. Let's discuss the linear transformation that matrix A applies to the two unit vectors e_1 and e_2 .

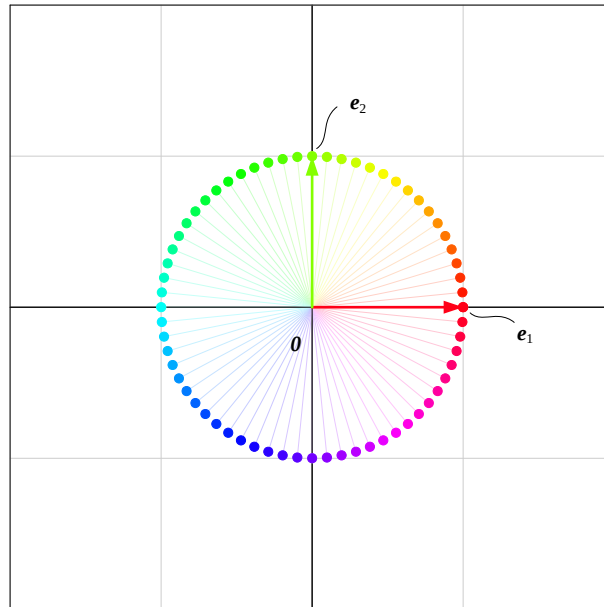


Figure 2. A Set of Unit Vectors

As Figure 3 shows, $\mathbf{A}$ acting on unit vector $\mathbf{e}_1$, gives $\mathbf{A}\mathbf{e}_1$

$$\mathbf{A}\mathbf{e}_1 = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/4 \\ -3/4 \end{bmatrix}$$

Compared with $\mathbf{e}_1$, $\mathbf{A}\mathbf{e}_1$ has changed in both magnitude and direction.

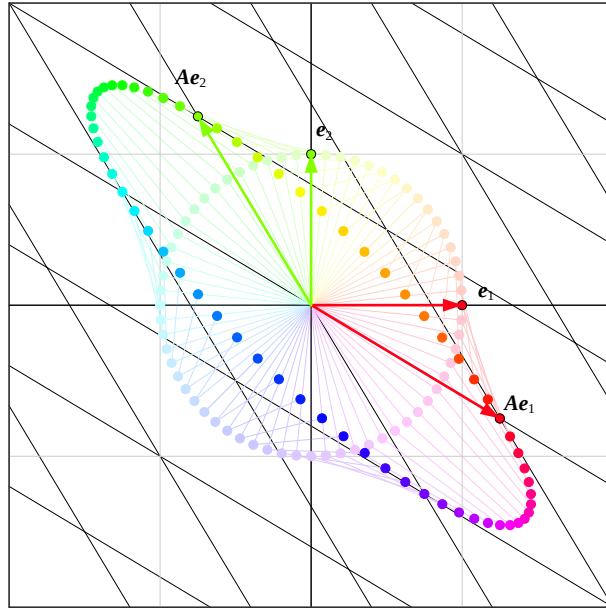
Likewise, $\mathbf{A}$ acting on unit vector $\mathbf{e}_2$, gives $\mathbf{A}\mathbf{e}_2$

$$\mathbf{A}\mathbf{e}_2 = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 5/4 \end{bmatrix}$$

Compared with $\mathbf{e}_2$, $\mathbf{A}\mathbf{e}_2$ has likewise changed in both magnitude and direction.

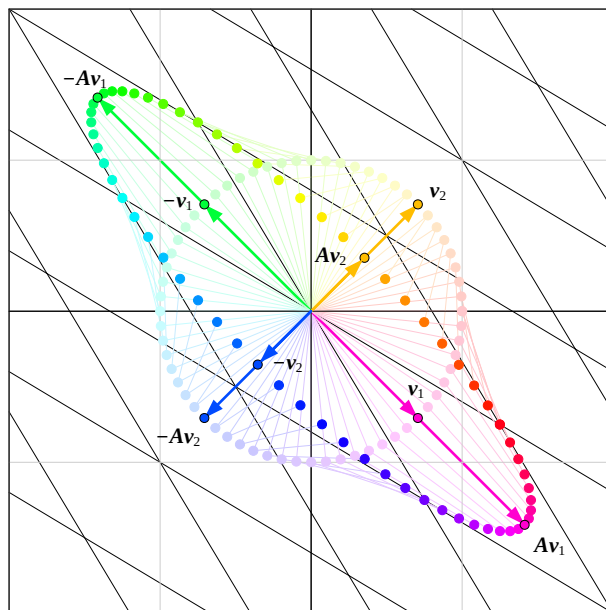
⚠ Note that the reason we focus on the standard orthonormal basis $[\mathbf{e}_1, \mathbf{e}_2]$ is that we want to see what parallelogram grid the unit square grid becomes, as well as the change in area before and after the transformation – that is, the change in the absolute value of the determinant.

? Please compute the determinant of $\mathbf{A}$.

Figure 3. Matrix A 's Linear Transformation of a Set of Unit Vectors

Eigenvectors

But looking carefully at Figure 4, we notice a few special vectors whose direction remains unchanged after being linearly transformed by matrix A – only their length gets scaled.

Figure 4. Matrix A 's Eigenvectors

For example, unit vector v_1 (magenta), after being linearly transformed by A , becomes

$$Av_1 = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} @ \underbrace{\begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}}_{v_1} = \begin{bmatrix} \sqrt{2} \\ -\sqrt{2} \end{bmatrix} = 2 \times \underbrace{\begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}}_{v_1}$$

That is to say

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$$A\mathbf{v}_1 = \underset{\lambda_1}{2}\mathbf{v}_1$$

This tells us that $\mathbf{v}_1$ is the eigenvector of A , with corresponding eigenvalue $\lambda_1 = 2$.

⚠ Note that eigenvalues and eigenvectors are matched in pairs.

Figure 4's, we also see that the opposite vector of $\mathbf{v}_1$ (magenta), $-\mathbf{v}_1$ (green), also satisfies

$$A@(-\mathbf{v}_1) = \underset{\lambda_1}{2}(-\mathbf{v}_1)$$

Furthermore, given a nonzero scalar k , the following equation always holds

$$A@(k\mathbf{v}_1) = \underset{\lambda_1}{2}(k\mathbf{v}_1)$$

The equation above tells us that the eigenvector corresponding to a given eigenvalue is not unique.

As Figure 5 shows, scaling the eigenvector by any nonzero scalar k still gives an eigenvector corresponding to the eigenvalue $\lambda_1 = 2$; therefore, all such eigenvectors form a line through the origin.

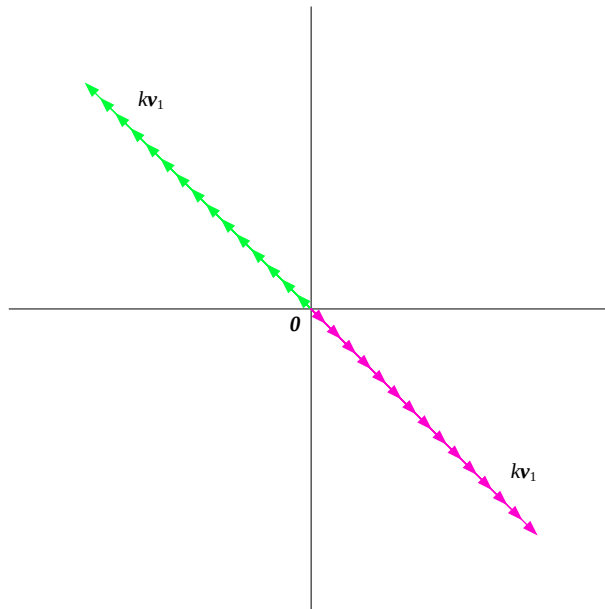


Figure 5. Eigenvectors Are Not Unique

⚠ Note that, unless stated otherwise, all eigenvectors in this book are normalized; that is, all eigenvectors in this book are direction vectors (unit vectors).

Figure 4's, we also notice another vector, $\mathbf{v}_2$ (orange), which after being linearly transformed by A becomes

$$A\mathbf{v}_2 = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} @ \underbrace{\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}}_{\mathbf{v}_2} = \begin{bmatrix} \sqrt{2}/4 \\ \sqrt{2}/4 \end{bmatrix} = \underset{\lambda_2}{0.5} \times \underbrace{\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}}_{\mathbf{v}_2}$$

That is to say

$$\mathbf{A}\mathbf{v}_2 = 0.5\mathbf{v}_2$$

This tells us that $\mathbf{v}_2$ is also an eigenvector of $\mathbf{A}$, with corresponding eigenvalue $\lambda_2 = 0.5$.



Please draw, on Figure 5, the line containing the eigenvectors corresponding to the eigenvalue $\lambda_2 = 0.5$.



You should notice that, in Figure 4's, the unit circle becomes a rotated ellipse! Why is that? What mathematical principle underlies this geometric transformation? This is a topic we'll discuss in Chapter 13 of this book.

Solving Eigenvalue Decomposition by Hand

Let's now talk about how to solve eigenvalue decomposition by hand.

Rearranging $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ into

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$$

Since $\mathbf{v}$ is a nonzero unit vector, after the linear transformation $\mathbf{A} - \lambda\mathbf{I}$ reduces the dimension, it yields the zero vector; therefore, the determinant of $\mathbf{A} - \lambda\mathbf{I}$ is 0, that is

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

For the matrix $\mathbf{A}$ in the equation above, $\mathbf{A} - \lambda\mathbf{I}$ is

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5/4 - \lambda & -3/4 \\ -3/4 & 5/4 - \lambda \end{bmatrix}$$

$\det(\mathbf{A} - \lambda\mathbf{I})$ corresponds to the following polynomial

$$\lambda^2 - \frac{5}{2}\lambda + 1$$

The equation above is called the characteristic polynomial.

Setting the characteristic polynomial to 0,

$$\lambda^2 - \frac{5}{2}\lambda + 1 = 0$$

we can solve for the eigenvalues.

As Figure 6 shows, if $b^2 - 4ac \geq 0$ is satisfied, the quadratic equation $ax^2 + bx + c = 0$ has two real roots

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

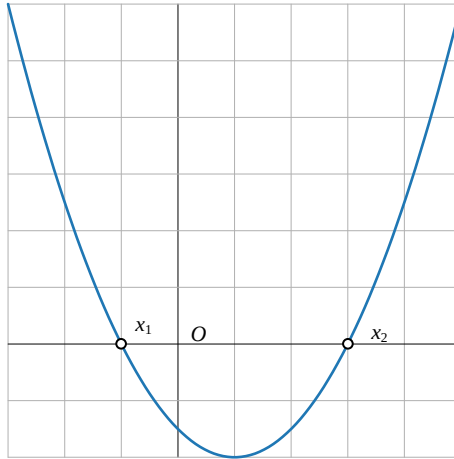


Figure 6. A Quadratic Function and the Roots of a Quadratic Equation

Based on this, matrix A 's two eigenvalues are

$$\lambda_1 = 2, \quad \lambda_2 = 1/2$$

For $\lambda_1 = 2$, let $\mathbf{v}_1 = [x_1, x_2]^T$. Substituting into the equation above, we get

$$\left(\begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) @ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{0}$$

This is exactly the homogeneous system of linear equations mentioned earlier in this book.

Let's briefly review: a homogeneous system of linear equations is one where all constant terms are zero; that is, in $\mathbf{Ax} = \mathbf{b}$, $\mathbf{b}$ is the zero vector, meaning the right-hand side of every equation is zero. Such a system always has at least one solution — the "zero solution," where all unknowns are zero, i.e., the zero vector. Besides the zero solution (the zero vector), a homogeneous system of linear equations may also have infinitely many nonzero solutions.

Rearranging the equation above, we get

$$\begin{bmatrix} -3/4 & -3/4 \\ -3/4 & -3/4 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{0}$$

Then we obtain the following equation

$$x_1 + x_2 = 0$$

As Figure 7 shows, any (nonzero) vector on this line through the origin is an eigenvector corresponding to the eigenvalue $\lambda_1 = 2$.

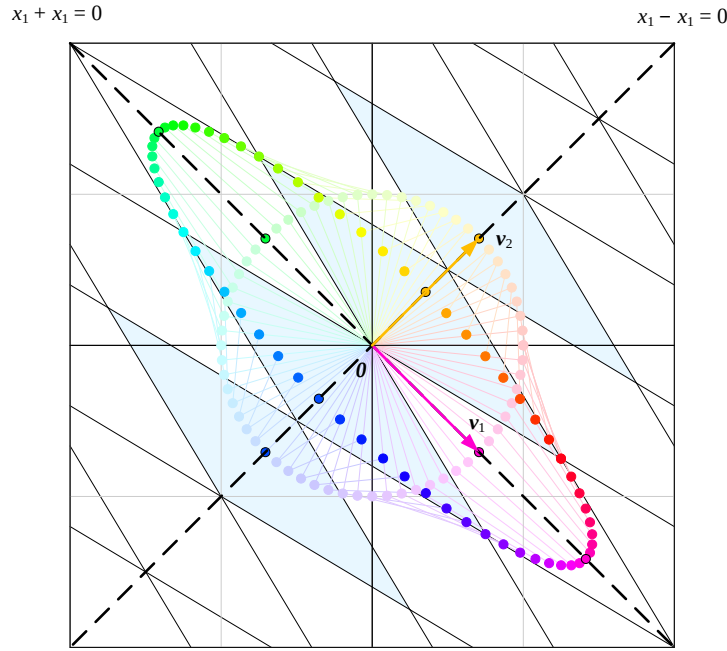


Figure 7. Two Lines (Through the Origin)

We choose the unit vector in the fourth quadrant as the eigenvector for eigenvalue

$$\mathbf{v}_1 = \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}$$

⚠ Note that we choose this $\mathbf{v}_1$ as the eigenvector so that $\mathbf{v}_1$ leads $\mathbf{v}_2$ counterclockwise (within 180 degrees); this way, the determinant of $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ is positive.

For $\lambda_2 = 0.5$, let $\mathbf{v}_2 = [x_1, x_2]^T$. Substituting into the equation above, we get

$$\left(\begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) @ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{0}$$

Rearranging the equation above, we get

$$\begin{bmatrix} 3/4 & -3/4 \\ -3/4 & 3/4 \end{bmatrix} @ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{0}$$

Then we obtain the following equation

$$x_1 - x_2 = 0$$

Any (nonzero) vector on this line through the origin is an eigenvector corresponding to the eigenvalue $\lambda_2 = 0.5$.

We choose the unit vector on the line in the first quadrant as $\mathbf{v}_2$

$$\mathbf{v}_2 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

As Figure 7 shows, have you noticed that $x_1 + x_2 = 0$ and $x_1 - x_2 = 0$ — these two lines pass through the origin, and happen to lie exactly at the intersection points of the parallelogram grid lines.

Written in matrix multiplication form

So far, we have obtained two matrix products

$$\begin{aligned} \mathbf{A}\mathbf{v}_1 &= \lambda_1\mathbf{v}_1 \\ \mathbf{A}\mathbf{v}_2 &= \lambda_2\mathbf{v}_2 \end{aligned}$$

Eigenvalue decomposition is a type of matrix decomposition (that is, writing a matrix as the product of several matrices); you might ask, why haven't we seen matrix decomposition take this particular form yet?

Using the third perspective on matrix multiplication again, let's take the two equations above and write them in sequence as

$$[\mathbf{A}\mathbf{v}_1 \quad \mathbf{A}\mathbf{v}_2] = [\lambda_1\mathbf{v}_1 \quad \lambda_2\mathbf{v}_2]$$

Rearranging further

$$\mathbf{A} \underbrace{[\mathbf{v}_1 \quad \mathbf{v}_2]}_{\mathbf{V}} = \underbrace{[\mathbf{v}_1 \quad \mathbf{v}_2]}_{\mathbf{V}} \underbrace{\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}}_{\mathbf{A}}$$

Recall, as introduced earlier in this book, that right-multiplying matrix $\mathbf{V}$ by a diagonal matrix scales the column vectors of matrix $\mathbf{V}$.

Therefore,

$$\mathbf{A}\mathbf{V} = \mathbf{V}\mathbf{A}$$

If matrix $\mathbf{V}$ has full rank ($\mathbf{v}_1, \mathbf{v}_2$ are linearly independent), $\mathbf{V}$ has an inverse matrix, and we can write $\mathbf{A}$ as

$$\mathbf{A} = \mathbf{V}\mathbf{A}\mathbf{V}^{-1}$$

At this point, we've arrived at the matrix decomposition form you've been wanting to see.

A matrix that can be written in the form shown above is called a diagonalizable matrix.

⚠ Note that for any real square matrix $\mathbf{A}$, the mere existence of eigenvalues and eigenvectors doesn't mean it can be diagonalized. As just mentioned, the eigenvectors being linearly independent is the key condition (that is, $\mathbf{V}$ is invertible). It's also worth noting that square matrix $\mathbf{A}$'s column vectors being linearly independent doesn't mean $\mathbf{A}$'s eigenvectors are linearly independent!

Rotate $\rightarrow$ Scale $\rightarrow$ Rotate

Substituting Figure 2's corresponding eigenvalues and eigenvectors into the equation above,

$$\mathbf{A} = \underbrace{\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}} \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\mathbf{V}^{-1}}$$

Have you noticed, to your surprise, that $\mathbf{V}$ in the equation above is exactly the rotation matrix we explained earlier in this book!

And more importantly, $\mathbf{V}$ is an orthogonal matrix, so

$$\mathbf{V}^{-1} = \mathbf{V}^T$$

That is to say, in this example, the geometric transformation resulting from A 's eigenvalue decomposition is "rotate $\rightarrow$ scale $\rightarrow$ rotate."

The reason this special transformation arises is that A is a symmetric matrix!

And the eigenvalue decomposition of a symmetric matrix has its own name — spectral decomposition!

This is the topic of our next section!

 Note that the eigenvalue decomposition of a non-symmetric matrix is not "rotate $\rightarrow$ scale $\rightarrow$ rotate"!

Computing Eigenvalue Decomposition Programmatically

Code 1 completes eigenvalue decomposition. Let's discuss the key lines below.

-  Use the `linalg.eig` function from NumPy to compute matrix A 's "eigenvalues" and "eigenvectors." Use `numpy.diag()` to turn the two eigenvalues extracted above into a diagonal matrix.
-  Extract the first eigenvalue from the list of eigenvalues.
-  Extract the first column from the eigenvector matrix V — that is, the first eigenvector. Note that the double square brackets `[:, [0]]` are used here to keep it in the form of a 2D array column vector.
-  And  perform the same operations for the second set of eigenvalues and eigenvectors.
-  Verify the eigenvalue decomposition result. Note that, in this example, V is invertible, so A can be diagonalized.

Code 1. Eigenvalue Decomposition |  LA_11_01_01.ipynb

```

## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 构造矩阵
A = np.array([[1.25, -0.75],
              [-0.75, 1.25]])

## 特征值分解
a lambda_s, V = np.linalg.eig(A)
  Lambda = np.diag(lambda_s)

## 第一组特征值、特征向量
# 特征值
b lambda_1 = lambda_s[0]
# 特征向量
c v_1 = V[:, [0]]
  A @ v_1
  lambda_1 * v_1

## 第二组特征值、特征向量
# 特征值
d lambda_2 = lambda_s[1]

# 特征向量
e v_2 = V[:, [1]]
  A @ v_2
  lambda_2 * v_2

## 特征值分解
f V @ Lambda @ np.linalg.inv(V)

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises in this section.

Q1. Compute the eigenvalue decomposition of the following matrices by hand, and verify your results using Python.

▶ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

▶ $\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

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▶ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

Q2. Identify which of the square matrices in Q1 can be diagonalized.